

TimeLightHub — Phase-1 Protocol v1.1: the single-cell experiment

Public summary for partner laboratories · Samer Beyrouty, Inventor & Architect · Connect@TimeLightHub.com · ORCID 0009-0003-8565-2605 · Zenodo DOI 10.5281/zenodo.22649324 · Patent Pending — GB2615417.9 (filed 18 June 2026) · **v1.1, 4 October 2026** (read step revised: the earlier 405 nm read is replaced by an off-resonance phase read with an explicit read-disturb measurement)

1. Purpose

TimeLightHub is a memory-centric photonic computing architecture whose energy claim reduces to one measurable quantity: **the energy to read one bit stored in a light-switchable (photochromic) memory cell, without disturbing it**. Phase 1 builds a single such cell and measures that quantity, together with the cell's readout latency and its read-disturb. The result is publishable on its own, whatever it shows, and it decides whether the rest of the architecture is worth building.

The experiment in one line. Fabricate one diarylethene (DAE) memory cell in a transparent polymer host, **write near 375 nm** (open → closed isomer), **reset at 532 nm** (closed → open), and **read off resonance**, at a wavelength neither isomer absorbs, through the refractive-index difference between isomers; then measure **(i) readout latency**, **(ii) energy per bit** for write, read and reset, and **(iii) read-disturb**.

2. The cell

A thin film of a diarylethene photochromic compound dispersed in a polymer host (SU-8 or PMMA class; the host and DAE derivative to be chosen with the partner from what the lab already handles). Stage A characterises the film on a planar substrate; Stage B repeats the readout inside a simple channel waveguide, with the doped section in one arm of a Mach-Zehnder, so the cell is probed the way the architecture will use it. No custom photonic process is required for Stage A; Stage B fits an existing polymer-waveguide or visible-wavelength SiN platform. Write and reset light can be delivered from above.

3. Measurements

Quantity	How it is obtained	Why it matters
M1 · Readout latency	Pulsed probe in two regimes: R1 off resonance (phase read, roughly 700–850 nm or 1310/1550 nm, chosen from the measured spectra) and R2 in the tail of the closed-form band (absorption read, about 630–680 nm). Report the shortest pulse that resolves the two states at a stated signal-to-noise ratio.	Sets the clock of the memory bank and of the AI engine's weight reads.
M2 · Switching energy per bit	Write and reset: minimum optical dose (pulse energy × spot area → energy per cell) that drives a full, repeatable state change. Read: probe energy per resolved read, for R1 and R2, at the signal-to-noise ratio used in M1, stated at the detector, launched and wall-plug.	The deciding number. Public energy window: tenths of a pJ per read = large advantage; a few pJ narrows it; above ten it disappears.
M3 · Read-disturb	Loss of stored state per unit of probe dose: the accumulated probe energy at which the contrast (R2) or the phase difference (R1) falls by 10%, converted to equivalent reads, with a 405 nm absorption read as the worst-case reference.	Decides whether the memory is truly non-volatile under use, or needs refreshing.
Supporting	Index difference between isomers versus wavelength, $\Delta n(\lambda)$; absorption of both isomers; cycle endurance; dark retention over days.	Confirms the cell is a usable non-volatile latch, not only a switch.

4. Method outline

- **Prepare** the DAE-doped film (spin-coat; thickness and loading chosen with the partner).
- **Characterise** absorption spectra and $\Delta n(\lambda)$ of open and closed states; choose the read wavelengths R1 and R2 from them.
- **Write** with 375 nm pulses of increasing dose; record the dose–response curve to saturation.
- **Read** in R1 and R2 with pulses of decreasing energy and duration until the states no longer separate → M1 and the read part of M2; then repeat reads to obtain M3.
- **Reset** with 532 nm; repeat the cycle to obtain endurance; hold written cells in the dark for retention.
- **Report** M1, M2 and the supporting metrics with uncertainties; feed them into the system-level energy model.

5. Success gates, scope and roles

Gate	Outcome
Go	Repeatable write/read/reset; in R1 less than 10% change after a probe dose equivalent to at least 10^{12} reads; read energy inside the public window, which is defined per weight read (one stored value, with the number of levels per cell stated), not per bit — proceed to Stage 2 (one byte and the first optical lookup).
Learn	Cell works but read energy is above the window, or only the low-disturb absorption read is usable — still publishable; the architecture's claims are revised with real numbers.
Stop	No repeatable switching, or no usable off-resonance index contrast, in the chosen host — reported openly.

Scope: months, not years; materials-scale, not program-scale; an ideal graduate project, scoped together with the partner. **Partner provides:** materials handling, film preparation, the write and reset sources (about 375 and 532 nm), one red or near-infrared probe, interferometric and time-resolved detection, and the student or staff time. **TimeLightHub provides:** the architecture and the public papers; under a simple mutual NDA, the engineering notes and a system-level simulator that shows how the measured numbers propagate to energy per answer; co-authorship on the resulting paper is expected and welcome.